УНИВЕРСИТЕТ ИННОПОЛИС

Semantic Search with BERT and TF-IDF

Abstract

Search engines often work well when the query uses the same words as the document.
But in real life, people can ask the same idea using different words.
In this project, we compare keyword and semantic search.

Experimental Setup

Python

and some libraries:
- Pandas
- Numpy
- Scikit-learn
- Spacy

Data

Dataset: CISI scientific article collection
Documents: 1460
Scored queries: 156

Methodology

Here i used different systems:
- **TF-IDF Search**. It gives high scores, when the query and document share important words.
Score(q, b) = cosine(tf-idf(q), tf-idf(b))
- **BERT Semantic Search**. It uses BERT for embeddings and find similar meaning even when exact words are different.
Score(q, b) = cosine(bert(q), bert(b))
- **Hybrid Fallback**. If query has too few words from TF-IDF, it switches to BERT.
If known_query_tokens / all_query_tokens < thr:
use BERT, otherwise use TF-IDF
- **Hybrid Weighted**. It combines TF-IDF score with BERT using some weight alpha.
Score = alpha * tf_idf + (1 - alpha) * BERT
- **Hybrid Interleave**. This method just uses Round-robin to take the same amount of items from each system.
If not in the list: append item, increase sys_pointer, and switch to the next system. Otherwise take next item.

Results

Query-level average

Method	Block MRR	Doc MRR	Doc Hit@10
Hybrid interleave	0.7957	0.8058	0.9744
BERT	0.7590	0.7952	0.9487
Hybrid weighted	0.7811	0.7897	0.9573
TF-IDF	0.7720	0.7733	0.9231
Hybrid fallback	0.7720	0.7733	0.9231

Group-wise query-level analysis

Group	Pairs	Queries	Method	Block MRR	Doc MRR	Doc Hit@10
Full	62	62	TF-IDF	0.9593	0.9593	1.0000
Full	62	62	Hybrid fallback	0.9593	0.9593	1.0000
Full	62	62	Hybrid weighted	0.9562	0.9562	0.9839
Full	62	62	Hybrid interleave	0.9534	0.9534	0.9839
Full	62	62	BERT	0.8283	0.8329	0.9677
None	318	36	BERT	0.0196	0.0209	0.0278
None	318	36	Hybrid interleave	0.0111	0.0119	0.0000
None	318	36	Hybrid weighted	0.0066	0.0073	0.0000
None	318	36	TF-IDF	0.0010	0.0012	0.0000
None	318	36	Hybrid fallback	0.0010	0.0012	0.0000
Partial	2011	57	BERT	0.6590	0.7285	0.9123
Partial	2011	57	Hybrid interleave	0.6011	0.6218	0.9474
Partial	2011	57	Hybrid weighted	0.5683	0.5858	0.9123
Partial	2011	57	TF-IDF	0.5496	0.5523	0.8421
Partial	2011	57	Hybrid fallback	0.5496	0.5523	0.8421

Conclusions

TF-IDF works well when the query and document use the same words.
BERT works better when the query uses different words but has the same meaning. **The best result comes from combining both methods**. The interleaving hybrid is simple and effective. It keeps strong keyword matches from TF-IDF and also adds semantic matches from BERT. **Overall, hybrid search is better than using only one method.**

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